

Practical 11 a.

```
niyathi@LAPTOP-NIYATHI:~/OSLab$ gcc practical11a.c -o practical11a
niyathi@LAPTOP-NIYATHI:~/OSLab$ ./practical11a
=====
      PRODUCER-CONSUMER PROBLEM
      Counting Semaphore Implementation
=====

Items produced/consumed : 100000
Number of producers    : 2
Number of consumers    : 2

Buffer Size      Time (seconds)      Throughput (items/s)
-----
5                0.759589             131650.22
10               0.401959             248781.80
20               0.184277             542662.15
50               0.070930             1409834.78

Experiment completed successfully.
```

Practical 11b.

```
niyathi@LAPTOP-NIYATHI:~/OSLab$ gcc practical11b.c -o practical11b -pthread
niyathi@LAPTOP-NIYATHI:~/OSLab$ ./practical11b
=====
      DEADLOCK DEMONSTRATION
=====

Thread 1: Trying to acquire Resource 1...
Thread 1: Acquired Resource 1.
Thread 2: Trying to acquire Resource 2...
Thread 2: Acquired Resource 2.
Thread 2: Trying to acquire Resource 1...
Thread 1: Trying to acquire Resource 2...
|
```